



Student Name: _____

Teacher's Name: _____

KIRRAWEE HIGH SCHOOL

2016

**YEAR 12
TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION**

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Board-approved calculators may be used
- Write your student name and teacher's name on the front of this examination paper and each answer booklet

Total marks – 100

Section I

Pages 3-5

10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II

Pages 6-13

90 marks

- Attempt Questions 11-16
- Allow about 2 hours and 45 minutes for this section

MC	Q11	Q12	Q13	Q14	Q15	Q16	TOTAL
/10	/15	/15	/15	/15	/15	/15	/100

Section I

Multiple Choice

10 marks

Answer on the Multiple Choice Answer Sheet provided

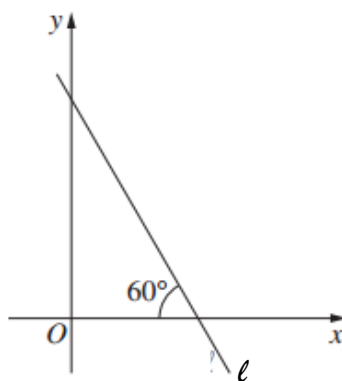
1 What is 4.097 84 correct to three significant figures?

- (A) 4.10 (B) 4.09 (C) 4.097 (D) 4.098

2 Which of the following is equal to $\frac{1}{2\sqrt{5}-\sqrt{3}}$?

- (A) $\frac{2\sqrt{5}-\sqrt{3}}{7}$ (B) $\frac{2\sqrt{5}+\sqrt{3}}{7}$ (C) $\frac{2\sqrt{5}-\sqrt{3}}{17}$ (D) $\frac{2\sqrt{5}+\sqrt{3}}{17}$

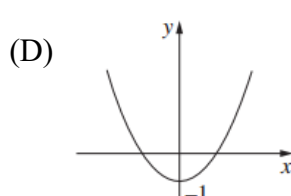
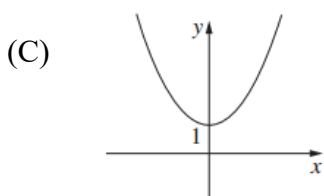
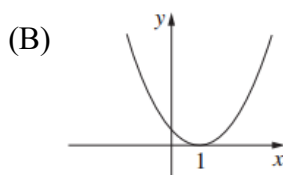
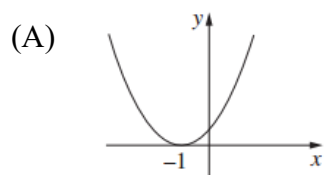
3 The diagram shows the line ℓ .



What is the slope of the line l ?

- (A) $-\sqrt{3}$ (B) $\sqrt{3}$ (C) $-\frac{1}{\sqrt{3}}$ (D) $\frac{1}{\sqrt{3}}$

4 Which graph best represents $y = (x-1)^2$?



- 5 Using the trapezoidal rule with 4 subintervals, which expression gives the approximate area under the curve $y = xe^x$ between $x = 1$ and $x = 3$?

- (A) $\frac{1}{4}(e^1 + 6e^{1.5} + 4e^2 + 10e^{2.5} + 3e^3)$
(B) $\frac{1}{4}(e^1 + 3e^{1.5} + 4e^2 + 5e^{2.5} + 3e^3)$
(C) $\frac{1}{2}(e^1 + 6e^{1.5} + 4e^2 + 10e^{2.5} + 3e^3)$
(D) $\frac{1}{2}(e^1 + 3e^{1.5} + 4e^2 + 5e^{2.5} + 3e^3)$

- 6 An infinite geometric series has a first term of 10 and a limiting sum of 12. What is the common ratio?

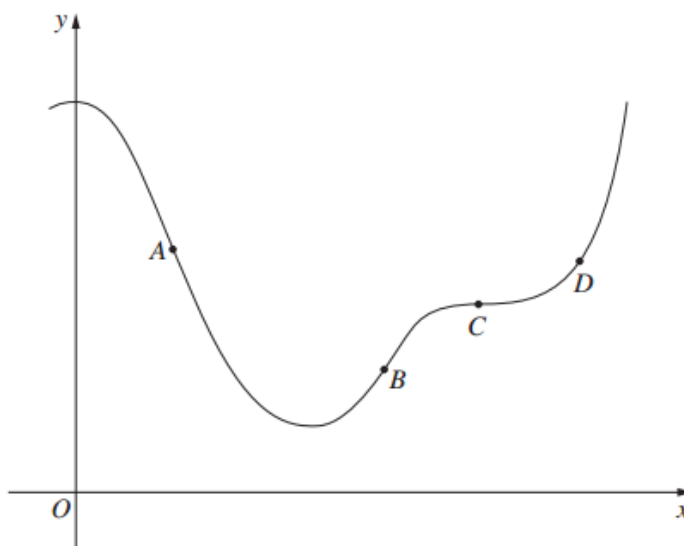
- (A) $\frac{1}{6}$ (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

- 7 A parabola has focus $(5,0)$ and directrix $x = 1$.

What is the equation of the parabola?

- (A) $y^2 = 16(x - 5)$
(B) $y^2 = -16(x - 5)$
(C) $y^2 = 8(x - 3)$
(D) $y^2 = -8(x - 3)$

- 8 The diagram shows points A , B , C and D on the graph $y = f(x)$



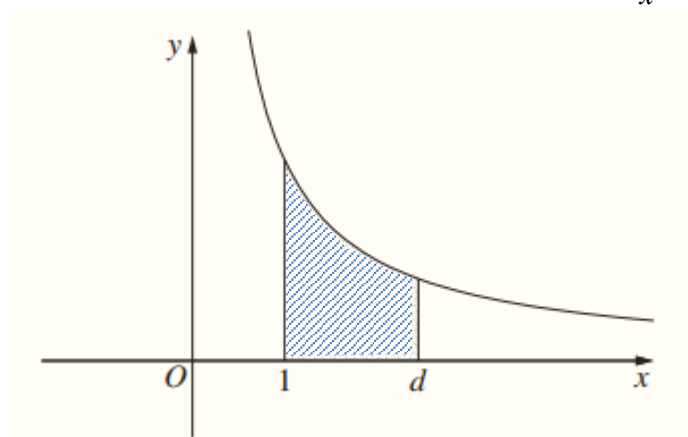
At which point is $f'(x) > 0$ and $f''(x) = 0$?

- (A) A (B) B (C) C (D) D

9 What are the solutions of $\sqrt{3} \tan x = -1$ for $0 \leq x \leq 2\pi$?

- (A) $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$
- (B) $\frac{2\pi}{3}$ and $\frac{5\pi}{3}$
- (C) $\frac{5\pi}{6}$ and $\frac{7\pi}{6}$
- (D) $\frac{5\pi}{6}$ and $\frac{11\pi}{6}$

10 The diagram shows the area under the curve $y = \frac{2}{x}$ from $x = 1$ to $x = d$



What values of d makes the shaded area equal to 2?

- (A) e^2
- (B) $e+1$
- (C) e
- (D) $2e$

Section II – Full Working Section

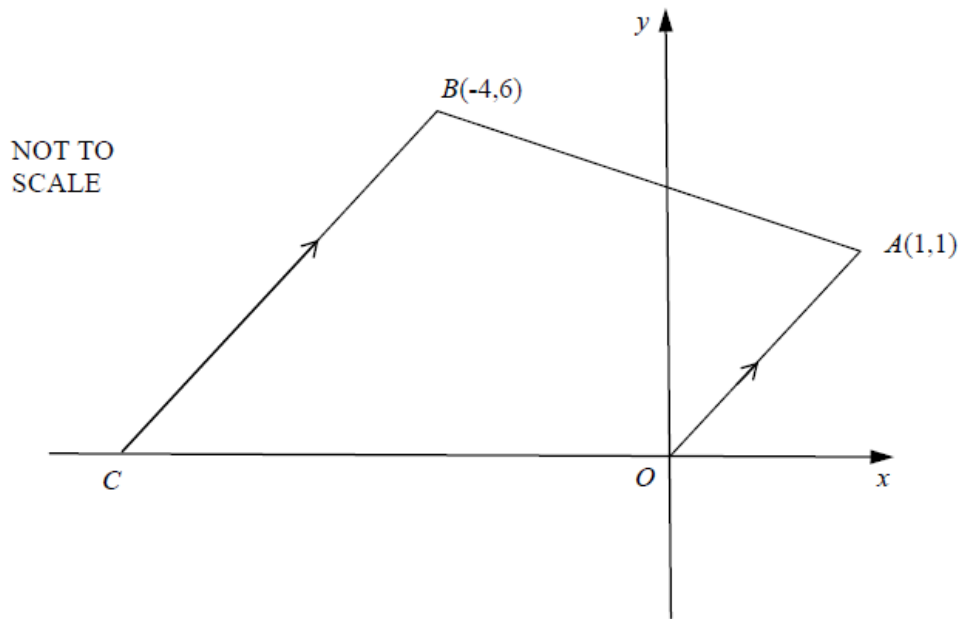
90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Question 11 (15 marks)	<u>Answer in the separate booklet provided</u>	Marks
(a) Solve $ 3x - 1 < 2$		2
(b) Factorise $1 - x^6$		2
(c) Differentiate $x^2 e^x$		2
(d) Expand and simplify $(\sqrt{3} - 1)(2\sqrt{3} + 5)$.		2
(e) Solve $2^{2x+1} = 32$		2
(f) Evaluate $\int_0^1 \frac{x^2}{x^3 + 1} dx$		3
(g) Evaluate $\int_0^{\frac{\pi}{4}} \cos 2x dx$		2

(a)

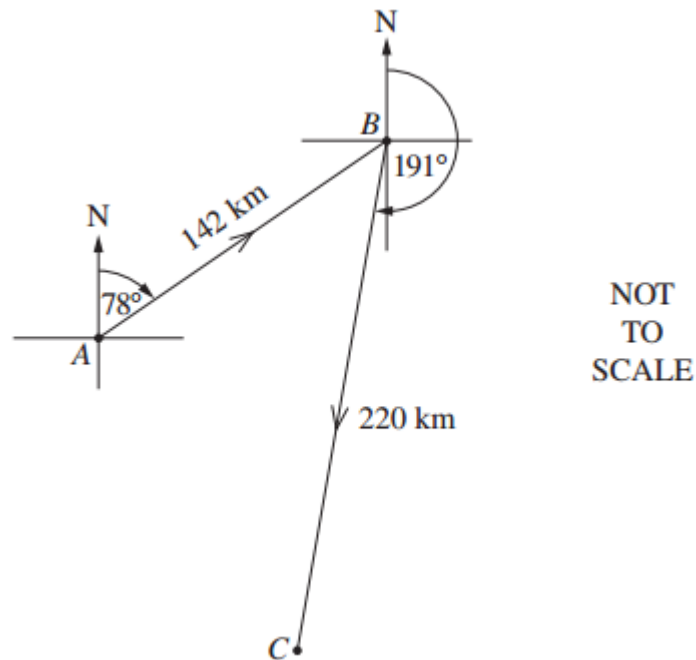


In the diagram, $OABC$ is a trapezium with $OA \parallel CB$. The coordinates of O , A and B are $(0,0)$, $(1,1)$ and $(-4,6)$ respectively.

- | | |
|---|---|
| (i) Calculate the length OA . | 1 |
| (ii) Write down the gradient of line OA . | 1 |
| (iii) What is the size of $\angle AOC$? | 1 |
| (iv) Find the equation of the line BC , and hence find the coordinates of C . | 2 |
| (v) Show that the perpendicular distance from O to the line BC is $5\sqrt{2}$. | 2 |
| (vi) Hence, or otherwise, calculate the area of the trapezium $OABC$. | 2 |
| | |
| (b) Evaluate the arithmetic series $2 + 5 + 8 + 11 + \dots + 1094$ | 2 |
| | |
| (c) Find $\int \frac{1}{(1+x)^2} dx$ | 2 |
| | |
| (d) The lengths of the sides of a triangle are 8 cm, 9 cm and 14 cm.
Find the size of the angle opposite the smallest side, to the nearest degree. | 2 |

Question 13 (15 marks)	<u>Answer in the separate booklet provided</u>	Marks
(a) (i) Differentiate $3 + \sin 2x$		2
(ii) Hence, or otherwise, find $\int \frac{\cos 2x}{3 + \sin 2x} dx$		2
(b) A function is defined by $f(x) = x^3 - 3x^2 - 9x + 10$.		
(i) Find the coordinates of the turning points and determine their nature.		3
(ii) Find the coordinates of the point of inflexion.		2
(iii) Sketch the graph for $y = f(x)$, showing all important information.		2
(c) Find the solutions of $2 \sin x = 1$ for $0 \leq x \leq 2\pi$.		2
(d) The area of a sector of a circle of radius 6 cm is 50 cm^2 . Find the length of the arc of the sector, correct to 2 decimal places.		2

- (a) Chris leaves island A in a boat and sails 142 km on a bearing of 078° to island B . Chris then sails on a bearing of 191° for 220 km to island C , as shown in the diagram.
(The diagram has also been replicated in your Question 14 Answer Booklet).

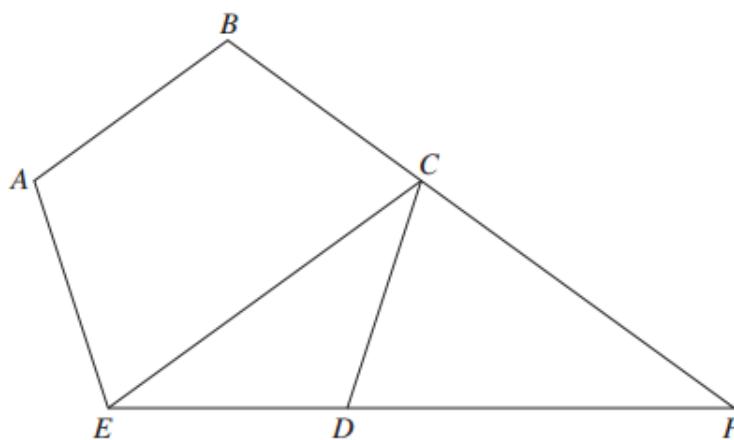


- | | |
|--|----------|
| (i) Show that the distance from island C to island A is approximately 210 km. | 2 |
| (ii) Chris wants to sail from island C directly to island A . On what bearing should Chris sail? Give your answer correct to the nearest degree. | 3 |
-
- (b) The roots of the equation $2x^2 - 5x + 12 = 0$ are α and β .
Find the value of:
- | | |
|--|----------|
| (i) $\alpha + \beta$ | 1 |
| (ii) $\alpha\beta$ | 1 |
| (iii) $\frac{1}{\alpha} + \frac{1}{\beta}$ | 2 |

Question 14 (continued)**Marks**

- (c) Sienna bought a new car costing \$19 600. The value of the car depreciates by 15% of the new price in its first year and at 12.5% of the previous year's value in each succeeding year. What is the value of the car (to the nearest dollar) after 8 years? **3**
- (d) Find the exact volume of the solid formed when the curve $y = \sqrt{x+3}$ is rotated about the x -axis between $x = 1$ and $x = 6$. **3**

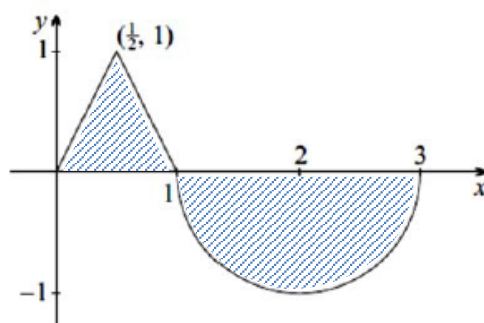
- (a) The diagram shows a regular pentagon $ABCDE$. Sides ED and BC are produced to meet at P .
(This diagram has been replicated in your Question 15 Answer Booklet)



In your answer booklet:

- (i) Find the size of $\angle CDE$ 2
- (ii) Hence, show that $\triangle EPC$ is isosceles 2
- (b) Find the equation of the locus of the point $P(x, y)$ that moves so that it is equidistant from the point $(2, 5)$ and the line $y = -3$. 2

(c)



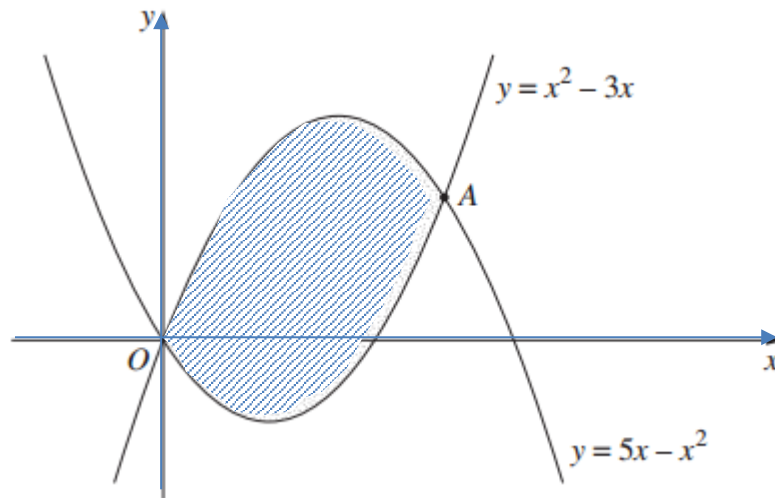
The diagram above shows the function $y = g(x)$ with domain $0 \leq x \leq 3$. 2

The arc is a semi-circle. Find $\int_0^3 g(x) dx$.

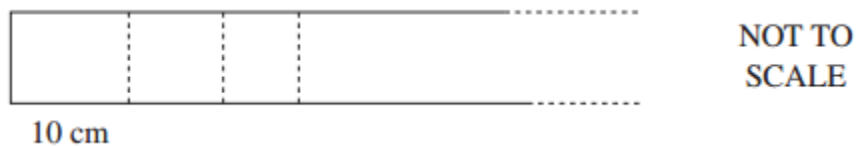
Question 15 (continued)

Marks

- (d) The diagram shows the parabola $y = 5x - x^2$ and $y = x^2 - 3x$. The parabolas intersect at the origin O and point A . The region between the two parabolas is shaded.



- (i) Find the x – coordinate of the point A . **1**
- (ii) Find the area of the shaded region, correct to 1 decimal place. **3**
- (e) Rectangles of the same height are cut from a strip and arranged in a row. The first rectangle has width 10 cm. The width of each subsequent rectangle is 96% of the width of the previous rectangle.



- (i) Find the length of the strip required to make the first ten rectangles. **2**
- (ii) Explain why a strip of length 3 m is sufficient to make any number of rectangles. **1**

- (a) The derivative of a function $f(x)$ is $f'(x) = 4x - 3$. The line $y = 5x - 7$ is tangent to the graph of $f(x)$. **3**

Find the function $f(x)$.

- (b) The graphs $y = x^2 - 3x - 5$ and $y = x + k$ have only one point of intersection, P .
- (i) Show that the x - coordinate of P satisfies $x^2 - 4x - 5 - k = 0$ **1**
- (ii) Find the value of k . **2**
- (iii) Find the coordinates of P . **2**

- (c) Andrew takes out a loan of \$360 000. The loan is to be repaid in equal monthly repayments, $\$M$, at the end of each month, over 25 years (300 months). Reducible interest is charged at 6% per annum, calculated monthly.

Let $\$A_n$ be the amount owing after the n th repayment.

- (i) Show that an expression for the amount owing after two months, $\$A_2$ is: **2**
- $$A_2 = 360\,000(1.005)^2 - M(1 + 1.005)$$
- (ii) Show that the monthly repayment is approximately \$2319.50. **2**

- (d) A soft drink manufacturer wants to minimise the amount of aluminium in its cans while still holding 375 mL of soft drink. Given that 375 mL has a volume of 375 cm³:

- (i) Show that the surface area of a can is given by $S = 2\pi r^2 + \frac{750}{r}$. **1**
- (ii) Find the radius of the can that gives the minimum surface area. **2**

End of paper.

Go back and check all of your answers.

Re-enter all of your calculations into your calculator and check that you still get the same answer.

2016 Trial (Mathematics) Solutions

Multiple Choice (1 mark each)

1. A
2. D
3. A
4. B
5. B
6. A
7. C
8. B
9. D
10. C

$$\begin{aligned} (d) & (\sqrt{3}-1)(2\sqrt{3}+5) \\ &= \sqrt{3}(2\sqrt{3}+5) - 1(2\sqrt{3}+5) \\ &= 6+5\sqrt{3}-2\sqrt{3}-5 \\ &= 1+3\sqrt{3} \quad \leftarrow (1) \end{aligned}$$

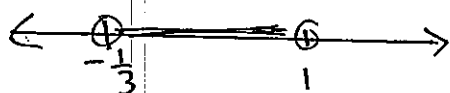
$$\begin{aligned} (e) & \begin{aligned} 2^{2x+1} &= 32 \\ 2^{2x+1} &= 2^5 \end{aligned} \quad \leftarrow (1) \\ & \begin{aligned} 2x+1 &= 5 \\ 2x &= 4 \\ x &= 2 \end{aligned} \quad \leftarrow (1) \end{aligned}$$

$$\begin{aligned} (f) & \int_0^1 \frac{x^2}{x^3+1} dx \\ & \frac{1}{3} \int_0^1 \frac{3x^2}{x^3+1} dx \quad \leftarrow (1) \\ &= \left[\frac{1}{3} \log_e(x^3+1) \right]_0^1 \quad \leftarrow (1) \\ &= \frac{1}{3} \log_e 2 - \frac{1}{3} \log_e 1 \\ &= \frac{1}{3} \log_e 2 - 0 \\ &= \frac{1}{3} \log_e 2 \quad \leftarrow (1) \\ &= 0.23104906 \end{aligned}$$

$$\begin{aligned} (g) & \int_0^{\frac{\pi}{4}} \cos 2x \cdot dx \\ &= \left[\frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{4}} \quad \leftarrow (1) \\ &= \left(\frac{1}{2} \sin \frac{\pi}{2} \right) - \left(\frac{1}{2} \sin 0 \right) \\ &= \frac{1}{2} \sin \frac{\pi}{2} \\ &= \frac{1}{2} \times 1 \\ &= \frac{1}{2} \quad \leftarrow (1) \end{aligned}$$

Qu. 11

$$\begin{aligned} (a) & |3x-1| < 2 \\ & \text{make } |3x-1| = 2 \\ & \therefore 3x-1=2 \text{ or } 3x-1=-2 \\ & 3x=3 \text{ or } 3x=-1 \\ & x=1 \text{ or } -\frac{1}{3} \quad \leftarrow (1) \end{aligned}$$



Test $x=0$

$$|3 \times 0 - 1| < 2$$

$$|-1| < 2$$

$$1 < 2 \checkmark$$

$$\therefore -\frac{1}{3} < x < 1 \quad \leftarrow (1) \quad \text{must have it written as 1 expression}$$

$$\begin{aligned} (b) & 1-x^6 \\ &= (1+x^3)(1-x^3) \quad \leftarrow (1) \\ &= (1+x)(1-x+x^2)(1-x)(1+x+x^2) \quad \leftarrow (1) \\ & \text{OR } (1-x^3)(1+x^2+x^4) \quad \leftarrow (1) \\ & (1+x)(1-x)(1+x^1+x^4) \quad \leftarrow (1) \end{aligned}$$

$$(c) y = x^2 e^x$$

$$y' = e^x \times 2x + x^2 e^x \quad \leftarrow (1)$$

$$y' = 2x e^x + x^2 e^x \quad \leftarrow (1)$$

$$y' = x e^x (2+x) \quad \leftarrow (1)$$

Qu. 12

$$(a) d = \sqrt{(1-0)^2 + (1-0)^2}$$

$$(i) d = \sqrt{1+1}$$

$$d = \sqrt{2} \leftarrow (1)$$

$$(ii) m = \frac{1-0}{1-0}$$

$$m = 1 \leftarrow (1)$$

$$(iii) \tan \theta = m$$

$$\tan \theta = 1$$

$$\theta = 45^\circ$$

$$\therefore \angle AOC = 180^\circ - 45^\circ = 135^\circ \leftarrow (1)$$

$$(iv) m = 1 \quad B(-4, 6)$$

$$y - 6 = 1(x + 4)$$

$$y - 6 = x + 4$$

$$x - y + 10 = 0 \leftarrow (1)$$

For point C, $y = 0$

$$x - 0 + 10 = 0$$

$$x = -10$$

$$\therefore C \text{ is } (-10, 0) \leftarrow (1)$$

$$(v) d = \frac{|1 \times 0 + (-1 \times 0) + 10|}{\sqrt{1^2 + (-1)^2}}$$

$$d = \frac{|10|}{\sqrt{2}} \leftarrow (1)$$

$$d = \frac{10}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$d = \frac{10\sqrt{2}}{2}$$

$$d = 5\sqrt{2} \leftarrow (1)$$

$$(vi) CB = \sqrt{(-4+10)^2 + (6-0)^2}$$

$$= \sqrt{36 + 36}$$

$$= \sqrt{72}$$

$$= 6\sqrt{2}$$

$$A = \frac{n}{2} (x+y)$$

$$A = \frac{5\sqrt{2}}{2} (6\sqrt{2} + \sqrt{2}) \leftarrow (1)$$

$$A = 30 + 5$$

$$A = 35 \text{ units}^2 \leftarrow (1)$$

$$(b) 1094 = 2 + (n-1) \times 3$$

$$1094 = 2 + 3n - 3$$

$$1094 = 3n - 1$$

$$3n = 1095$$

$$n = 365$$

$$\leftarrow (1)$$

$$\therefore S_{365} = \frac{365}{2} (2 + 1094)$$

$$= 200020 \leftarrow (1)$$

$$(c) \int \frac{1}{(1+x)^2} \cdot dx$$

$$= \int (1+x)^{-2} \cdot dx$$

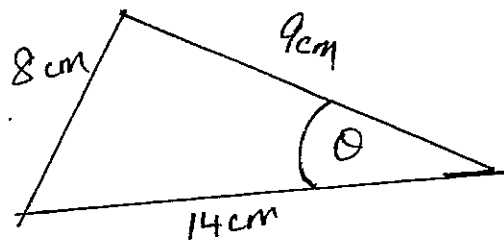
$$= \frac{(1+x)^{-1}}{-1 \times 1} + C \leftarrow (1)$$

$$= \frac{(1+x)^{-1}}{-1} + C$$

$$= -\frac{1}{(x+1)} + C \leftarrow (1)$$

* you could use 1 mark for "C"

(d)



$$\cos \theta = \frac{14^2 + 9^2 - 8^2}{2 \times 14 \times 9} \leftarrow (1)$$

$$\cos \theta = \frac{213}{252}$$

$$\theta = 32^\circ \leftarrow (1)$$

Qu. 13

(a)(i) $y = 3 + \sin 2x$ 1 for the 2
 $y' = 2 \cos 2x$ 1 for $\cos 2x$

(ii) $\frac{1}{2} \int \frac{2 \cos 2x}{3 + \sin 2x} \cdot dx \leftarrow (1)$
 $= \frac{1}{2} \log_e (3 + \sin 2x) + C \leftarrow (1)$

(b)(i) $f(x) = x^3 - 3x^2 - 9x + 10$
 $f'(x) = 3x^2 - 6x - 9$
 make $f'(x) = 0$
 $3x^2 - 6x - 9 = 0$
 $3(x^2 - 2x - 3) = 0$
 $3(x - 3)(x + 1) = 0$
 $x = 3, -1$
 when $x = 3, y = -17$
 when $x = -1, y = 15$
 $(3, -17)$ and $(-1, 15) \leftarrow (1)$

$f''(x) = 6x - 6$
 for $x = 3$
 $f''(3) = 6(3) - 6 = 12$
 since $f''(x) > 0$
 $(3, -17)$ is a minimum turning point (concave up) $\leftarrow (1)$

for $x = -1$
 $f''(-1) = 6(-1) - 6 = -12$
 since $f''(x) < 0$
 $(-1, 15)$ is a maximum turning point (concave down) $\leftarrow (1)$

(ii) make $f''(x) = 0$
 $6x - 6 = 0$
 $x = 1$

when $x = 1, y = -1$

$\therefore (1, -1) \leftarrow (1)$

test concavity

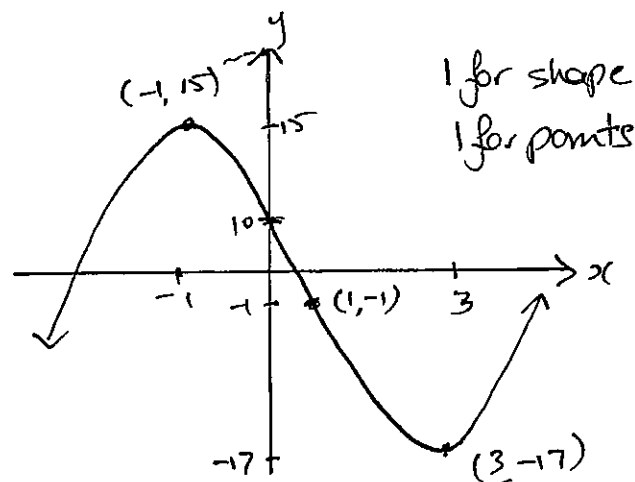
x	0	1	2
$f''(x)$	-6	0	6

when $x = 0$
 $f''(0) = -6$

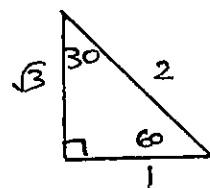
when $x = 2$
 $f''(2) = 6 \leftarrow (1)$

$\therefore$ since change in concavity
 $(1, -1)$ is a point of inflexion

(iii)



(c) $2 \sin x = 1$
 $\sin x = \frac{1}{2}$
 $\sin 30^\circ = \frac{1}{2}$



$x \quad S \quad A \quad x$
 $\frac{\quad}{T \quad C}$

$x = \frac{30\pi}{180}, \frac{150\pi}{180}$

$x = \frac{\pi}{6}, \frac{5\pi}{6} \leftarrow (1)$

(d) $A = \frac{1}{2} r^2 \theta$
 $50 = \frac{1}{2} \times 6^2 \times \theta$
 $50 = 18 \times \theta$

$\theta = \frac{50}{18} \leftarrow (1)$

$l = r\theta$

$l = 6 \times \frac{50}{18}$

$l = 16.67 \text{ cm} \leftarrow (1)$

Qu. 14

$$(a)(i) d^2 = 142^2 + 220^2 - 2(142)(220)\cos 67^\circ \leftarrow (1)$$

$$d^2 = 144151.11909$$

$$\therefore d \approx 210 \text{ km} \leftarrow (1)$$

$$(ii) \cos C = \frac{210^2 + 220^2 - 142^2}{2 \times 210 \times 220} \leftarrow (1)$$

$$\cos C = \frac{72336}{92400}$$

$$C = 38^\circ$$

$$\text{bearing is } 360^\circ - 27^\circ \leftarrow (1)$$

$$= 333^\circ \text{ T} \leftarrow (1)$$

$$(b) 2x^2 - 5x + 12 = 0$$

$$(i) \alpha + \beta = -\frac{b}{a}$$

$$= \frac{5}{2} \leftarrow (1)$$

$$(ii) \alpha\beta = \frac{c}{a}$$

$$= \frac{12}{2}$$

$$= 6 \leftarrow (1)$$

$$(iii) \frac{1}{\alpha} + \frac{1}{\beta}$$

$$= \frac{\alpha + \beta}{\alpha\beta} \leftarrow (1)$$

$$= \frac{5}{2} \div 6$$

$$= \frac{5}{12} \leftarrow (1)$$

$$(c) 19600 \times 0.25 \leftarrow (1)$$

$$= \$16660$$

$$A = 16660(1 - 0.125)^7 \leftarrow (1)$$

$$A = 16660(0.875)^7$$

$$A = \$6542.31 \leftarrow (1)$$

$$(d) V = \pi \int_1^6 (\sqrt{x+3})^2 \cdot dx$$

$$V = \pi \int_1^6 (x+3) \cdot dx \leftarrow (1)$$

$$V = \pi \left[\frac{x^2}{2} + 3x \right]_1^6 \leftarrow (1)$$

$$V = \pi \left[(18+18) - \left(\frac{1}{2} + 3 \right) \right]$$

$$V = \pi \left[36 - 3\frac{1}{2} \right]$$

$$V = \frac{65\pi}{2} \text{ units}^3 \leftarrow (1)$$

- Qu. 15

(i) Angle size = $\frac{(n-2) \times 180^\circ}{n}$

$$\angle CDE = \frac{3 \times 180^\circ}{5} = 108^\circ \quad \leftarrow (1)$$

(ii) $ED = DC$ (equal sides of a regular pentagon)

$\therefore \angle DEC = \angle DCE = 36^\circ$ (base angles of an isosceles triangle)

$$\angle BCD = 108^\circ$$

$$\angle BCE = 72^\circ (108^\circ - 36^\circ)$$

$$\angle DCP = 72^\circ \text{ (angles on a straight line add to } 180^\circ) \quad \leftarrow (1)$$

$$\angle EDC = \angle DCP + \angle DPC \text{ (exterior angle theorem of a triangle)}$$

$$108 = 72 + \angle DPC$$

$$\angle DPC = 36^\circ$$

$\therefore \angle CED = \angle CPD = 36^\circ$ (base angles of an isosceles triangle)

$\therefore \triangle EPC$ is isosceles $\leftarrow (1)$

(b) $PA = PB$

$$\sqrt{(x-2)^2 + (y-5)^2} = \sqrt{(x-x)^2 + (y+3)^2} \quad \leftarrow (1)$$

$$x^2 - 4x + 4 + y^2 - 10y + 25 = y^2 + 6y + 9$$

$$x^2 - 4x - 16y + 20 = 0 \quad \leftarrow (1)$$

(c) $\int_0^3 g(x) \cdot dx$

$$= \frac{bh}{2} - \frac{\pi r^2}{2} \quad \leftarrow (1)$$

$$= \frac{1 \times 1}{2} - \frac{\pi \times 1^2}{2}$$

$$= \frac{1}{2} - \frac{\pi}{2} \quad \leftarrow (1)$$

(d)(i) $x^2 - 3x = 5x - x^2$

$$2x^2 - 8x = 0$$

$$2x(x-4) = 0$$

$$x=0, x=4$$

$\therefore x$ coordinate of A is 4 $\leftarrow (1)$

(ii) $A = \int_0^4 5x - x^2 - (x^2 - 3x) \cdot dx$

$$A = \int_0^4 5x - x^2 - x^2 + 3x \cdot dx \quad \leftarrow (1)$$

$$A = \int_0^4 8x - 2x^2 \cdot dx$$

$$A = \left[\frac{8x^2}{2} - \frac{2x^3}{3} \right]_0^4 \quad \leftarrow (1)$$

$$A = \left[4x^2 - \frac{2x^3}{3} \right]_0^4$$

$$A = (64 - 42\frac{2}{3}) - (0)$$

$$A = 21.3 \text{ units}^2 \quad \leftarrow (1)$$

(e)(i) $a=10 \quad r=0.96$

$$S_{10} = \frac{10(1-0.96^{10})}{1-0.96} \quad \leftarrow (1)$$

$$= \frac{10(1-0.96^{10})}{0.04}$$

$$= 83.791841 \text{ cm} \quad \leftarrow (1)$$

(ii) $3m = 300 \text{ cm}$

$$S = \frac{a}{1-r}$$

$$S = \frac{10}{1-0.96}$$

$$S = \frac{10}{0.04}$$

$$S = 250 \quad \leftarrow (1)$$

Since the limiting sum is 250 and less than 300, 3m is sufficient to make any number of rectangles

Qu. 1b

$$1) f'(x) = 4x - 3$$

$$\text{and } m = 5$$

$$\therefore 4x - 3 = 5$$

$$4x = 8$$

$$x = 2$$

$$\text{when } x = 2$$

$$y = 5(2) - 7$$

$$y = 3$$

$$\therefore (2, 3) \in (1)$$

$$f'(x) = 4x - 3$$

$$f(x) = \frac{4x^2}{2} - 3x + C$$

$$= 2x^2 - 3x + C \in (1)$$

$$\text{Substn } (2, 3)$$

$$3 = 2(2)^2 - 3(2) + C$$

$$3 = 8 - 6 + C$$

$$3 = 2 + C$$

$$C = 1$$

$$\therefore f(x) = 2x^2 - 3x + 1 \in (1)$$

$$(b)(i) y = x^2 - 3x - 5$$

$$y = x + k$$

$$x^2 - 3x - 5 = x + k$$

$$x^2 - 4x - 5 - k = 0 \in (1)$$

(ii) For 1 point of intersection

$$\Delta = 0$$

$$\Delta = b^2 - 4ac$$

$$(-4)^2 - (4 \times 1 \times [-5 - k]) = 0 \in (1)$$

$$16 - 4(-5 - k) = 0$$

$$16 + 20 + 4k = 0$$

$$4k = -36$$

$$k = -9 \in (1)$$

$$(iii) x^2 - 4x - 5 + 9 = 0$$

$$x^2 - 4x + 4 = 0$$

$$(x - 2)(x - 2) = 0$$

$$\therefore x = 2 \in (1)$$

$$\text{when } x = 2$$

$$y = x + k$$

$$y = 2 + -9$$

$$y = -7$$

$$\therefore P_{10}(2, -7) \in (1)$$

$$(c)(i) r = \frac{6}{100} \div 12$$

$$= 0.005$$

$$A_1 = 360\,000(1.005)^1 - M$$

$$A_2 = A_1(1.005)^1 - M \in (1)$$

$$A_2 = [360\,000(1.005)^1 - M](1.005)^1 - M$$

$$A_2 = 360\,000(1.005)^2 - M(1.005)^1 - M$$

$$A_2 = 360\,000(1.005)^2 - M(1.005 + 1)$$

$$A_2 = 360\,000(1.005)^2 - M(1 + 1.005) \in (1)$$

$$(ii) A_{300} = 0$$

$$A_{300} = 360\,000(1.005)^{300} - M(1 + 1.005 + \dots + 1.005^{299})$$

$$0 = 360\,000(1.005)^{300} - M(1 + 1.005 + \dots + 1.005^{299})$$

$$M(1 + 1.005 + \dots + 1.005^{299}) = 360\,000(1.005)^{300}$$

$$M = 360\,000(1.005)^{300} \div \frac{1(1.005^{300} - 1)}{1.005 - 1}$$

$$M = 360\,000(1.005)^{300} \times \frac{0.005}{(1.005^{300} - 1)}$$

$$M = \$2319.485045$$

$$\therefore M = \$2319.50 \in (1)$$

$$(d) V = \pi r^2 h$$

$$S.A = 2\pi r^2 + 2\pi rh$$

$$(i) 375 = \pi r^2 h$$

$$h = \frac{375}{\pi r^2}$$

$$S = 2\pi r^2 + 2\pi r \times \frac{375}{\pi r^2}$$

$$S = 2\pi r^2 + \frac{750}{r} \in (1)$$

$$(ii) S = 2\pi r^2 + 750r^{-1}$$

$$S' = 4\pi r - 750r^{-2}$$

$$S' = 0$$

$$4\pi r - \frac{750}{r^2} = 0$$

$$4\pi r^3 - 750 = 0 \in (1)$$

$$4\pi r^3 = 750$$

$$r^3 = \frac{750}{4\pi}$$

$$r = \sqrt[3]{\frac{750}{4\pi}}$$

$$r = \sqrt[3]{59.68310366}$$

$$r = 3.907963209 \in (1)$$